



INFRASTRUCTURE

Load Balancer-F5

Document

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VERSION HISTORY

Version	Date	Name	Comments/Changes
1.0	06-24-20	Shilpa Kaul	Created initial draft of the document.
1.0	07-09-20	xxx Shilpa Kaul	Added diagrams to the High-Level Diagram – MDC, LAS. Added diagram to the Load Balancing Services - Lennar.com
1.0	09-10-20	xxx	Added description and diagrams to the following sections: • Splunk and Syslog Environment • ESB Services • Divisions Micro Websites • LDAP • Lennar.com Environment (Dev)
1.0	09-14-20	xxx	Reviewed the document.
1.0	09-22-20	Shilpa Kaul	Released the document.

DOCUMENT APPROVER

Document Approver	Designation	Signature	Approval Date
xxx	Lead System Engineer, Network, Technology Operations		

CONTRIBUTORS

Contributor	Designation
xxx	Lead System Engineer, Network, Technology Operations
xxx	Sr System Engineer, Network, Technology Operations

ABBREVIATIONS AND ACRONYMS

Term	Description
ESB	Enterprise Service Bus
SOA	Service Oriented Architecture

OVERVIEW

This document describes the network environment and infrastructure at Lennar, this includes our datacenters and corporate offices. Provides essential information regarding tier design at Lennar, configuration and hardware standards, and cloud infrastructure.

AUDIENCE

The intended audience of this document are the Lennar Technology Solutions (LTS) personnel.

HIGH LEVEL DIAGRAM

MDC

The diagram provides physical network connectivity details of the Miami datacenter F5 i5800 series appliances. This diagram also reflects the layer 2 information that each physical connection is configured for.

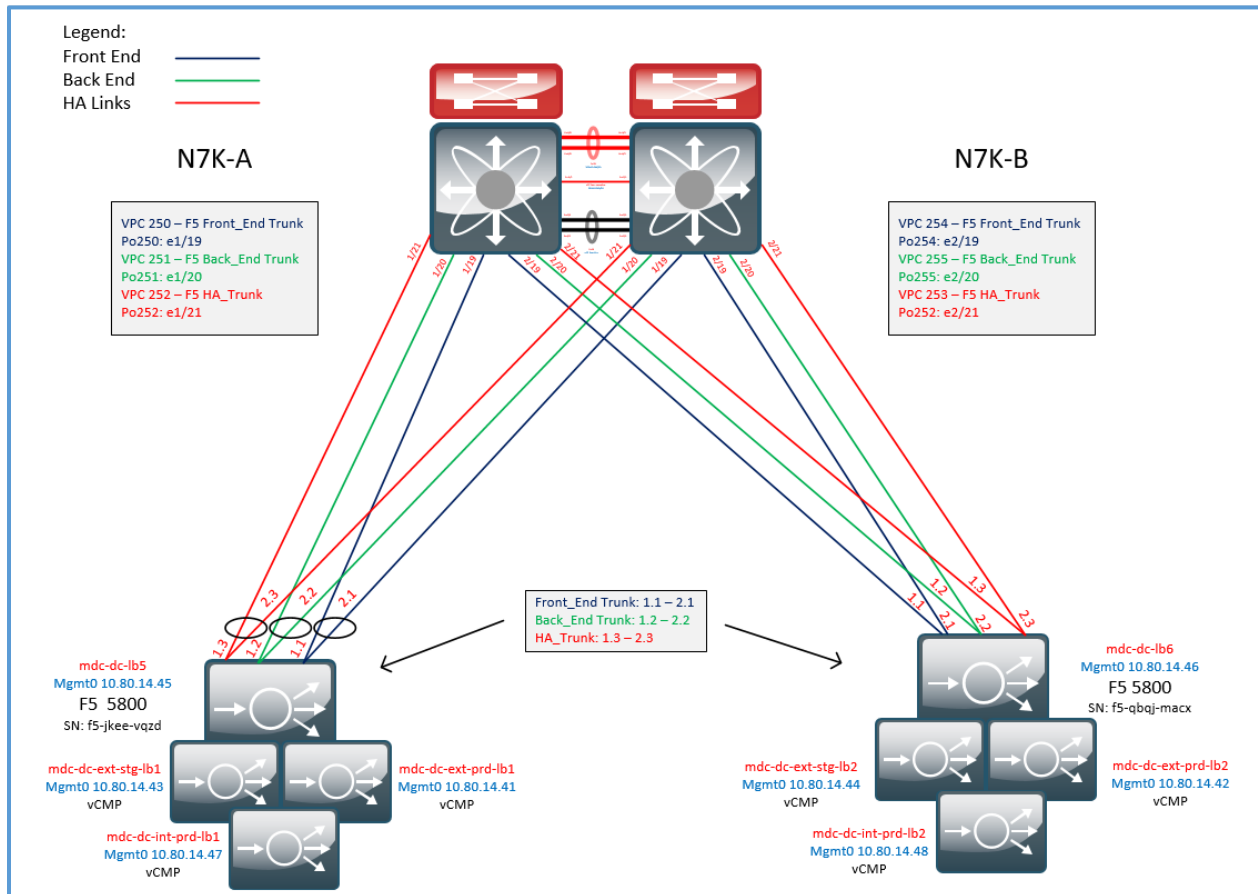


Figure 1: MDC - High Level Diagram

LAS

The diagram provides physical network connectivity details of the Las Vegas datacenter F5 i5800 series appliances. This diagram also reflects the layer 2 information that each physical connection is configured for.

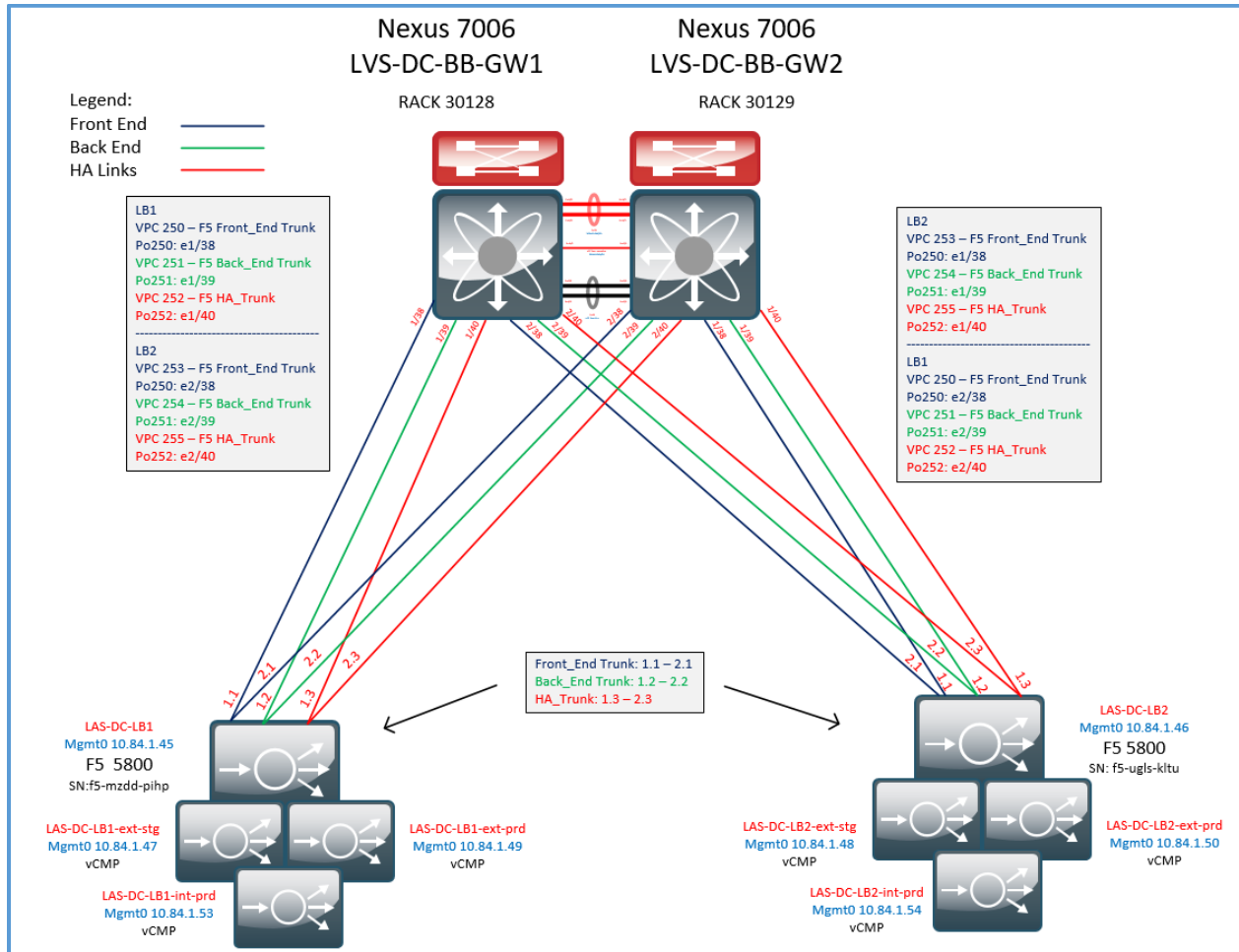


Figure 2: LAS - High Level Diagram

LOAD BALANCING SERVICES

This section represents the environment, load balancers and services used at Lennar.

EXTERNAL

Lennar.com Environment (PROD)

The diagram reflects the client to server connectivity details for accessing the lennar.com application environment. In this diagram there are different components of the lennar.com application and the F5 server pool that is used to access that component.

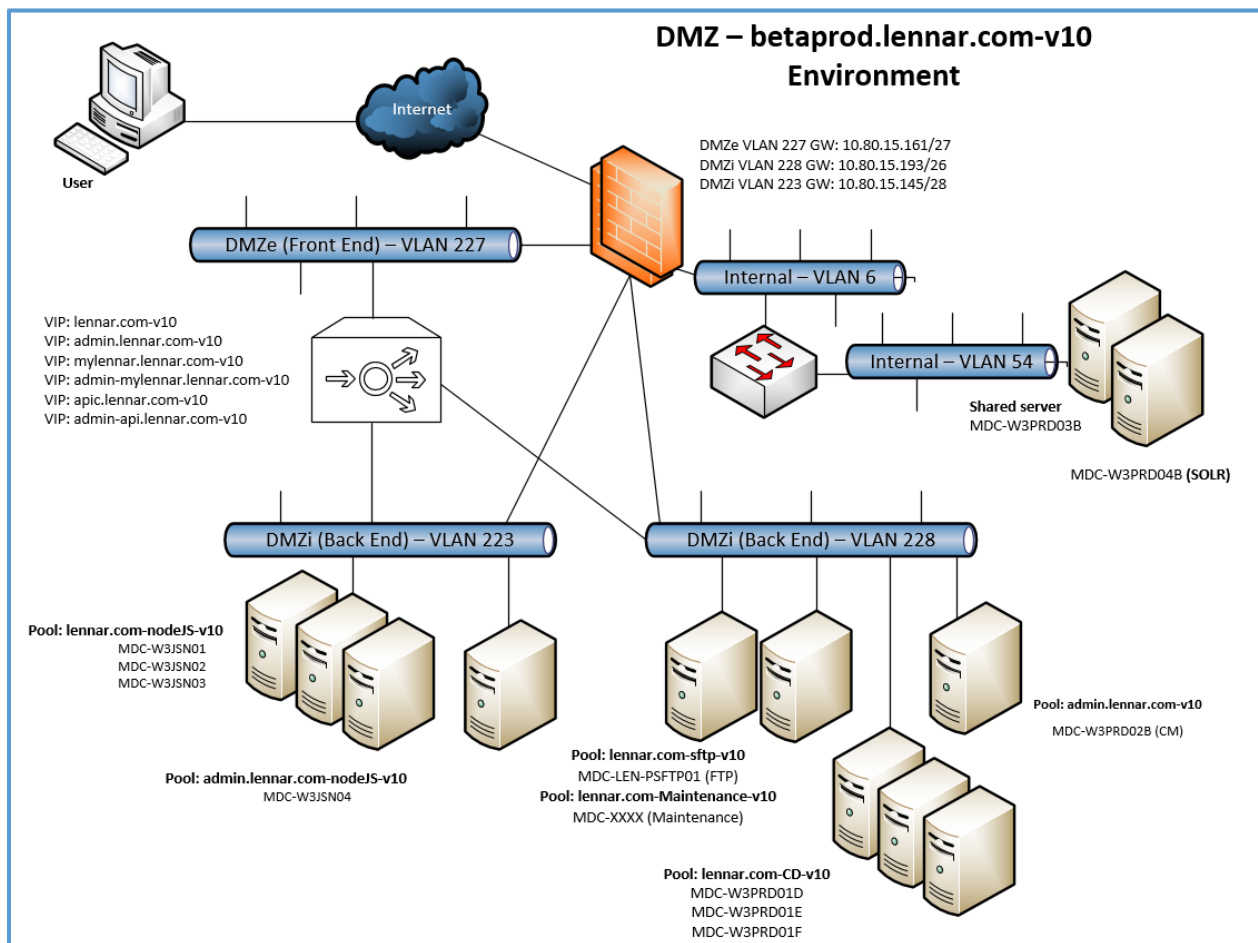


Figure 3: Lennar.com Environment

Splunk and Syslog Environment

The MDC and LAS environments both service the Splunk and Syslog applications. Both environments leverage the F5 appliances to service a single application across multiple backend servers.

Below are the environment specific details.

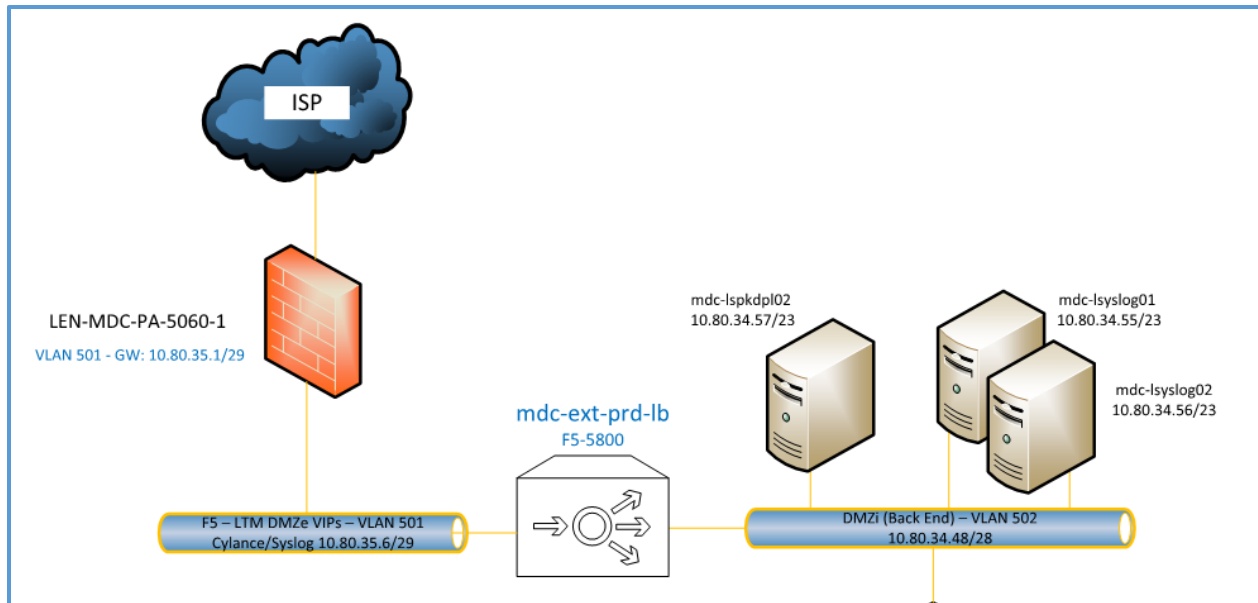


Figure 4: MDC – Splunk and Syslog Environment

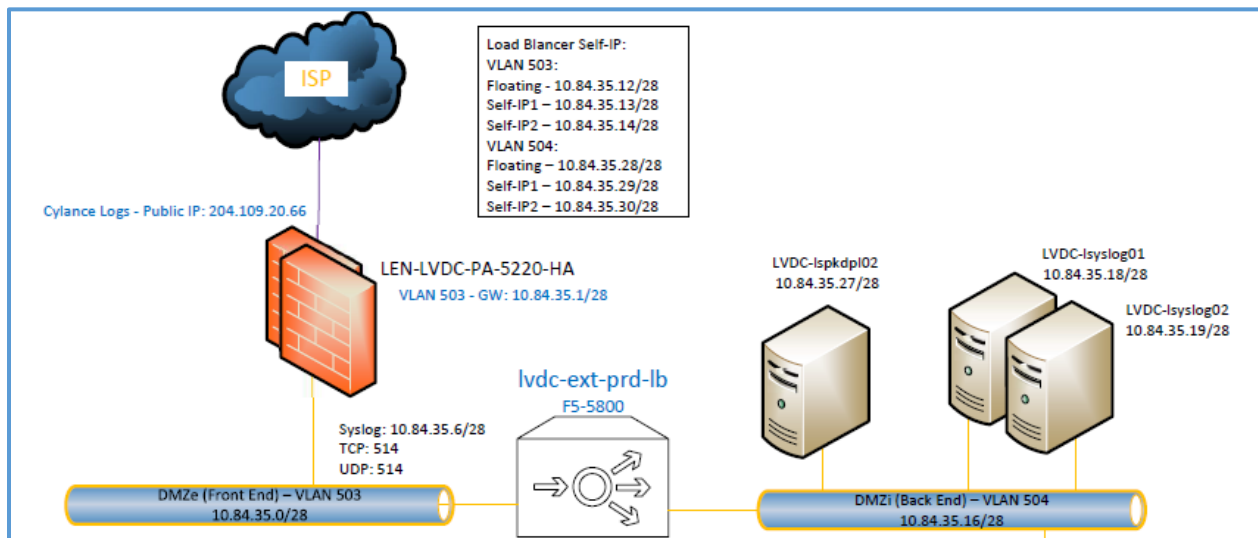


Figure 5: LAS – Splunk and Syslog Environment

Enterprise Service Bus (ESB) Services

Enterprise Service Bus (ESB) services is a software architecture for distributed computing and uses the general client-server model to facilitate a communication system between mutually interacting software applications in a SOA. The diagram below shows the connectivity to the ESB application. The external connection uses the HTTP connection that is redirected to HTTPS and the internal ESB application uses the HTTP protocol to host the application. This traffic is load balanced to application servers via the AWS VPN Gateway.

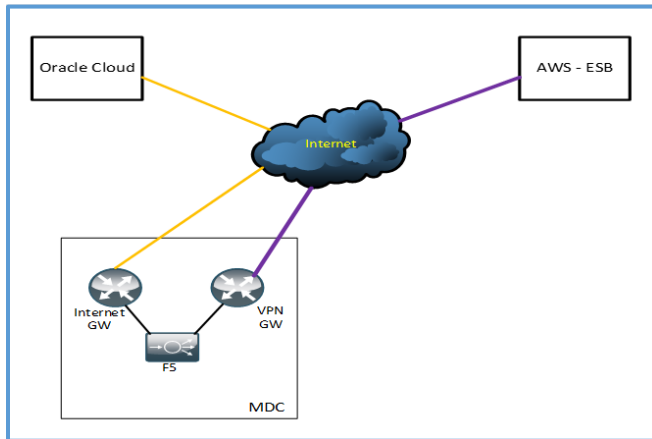


Figure 6: Enterprise Service Bus Services

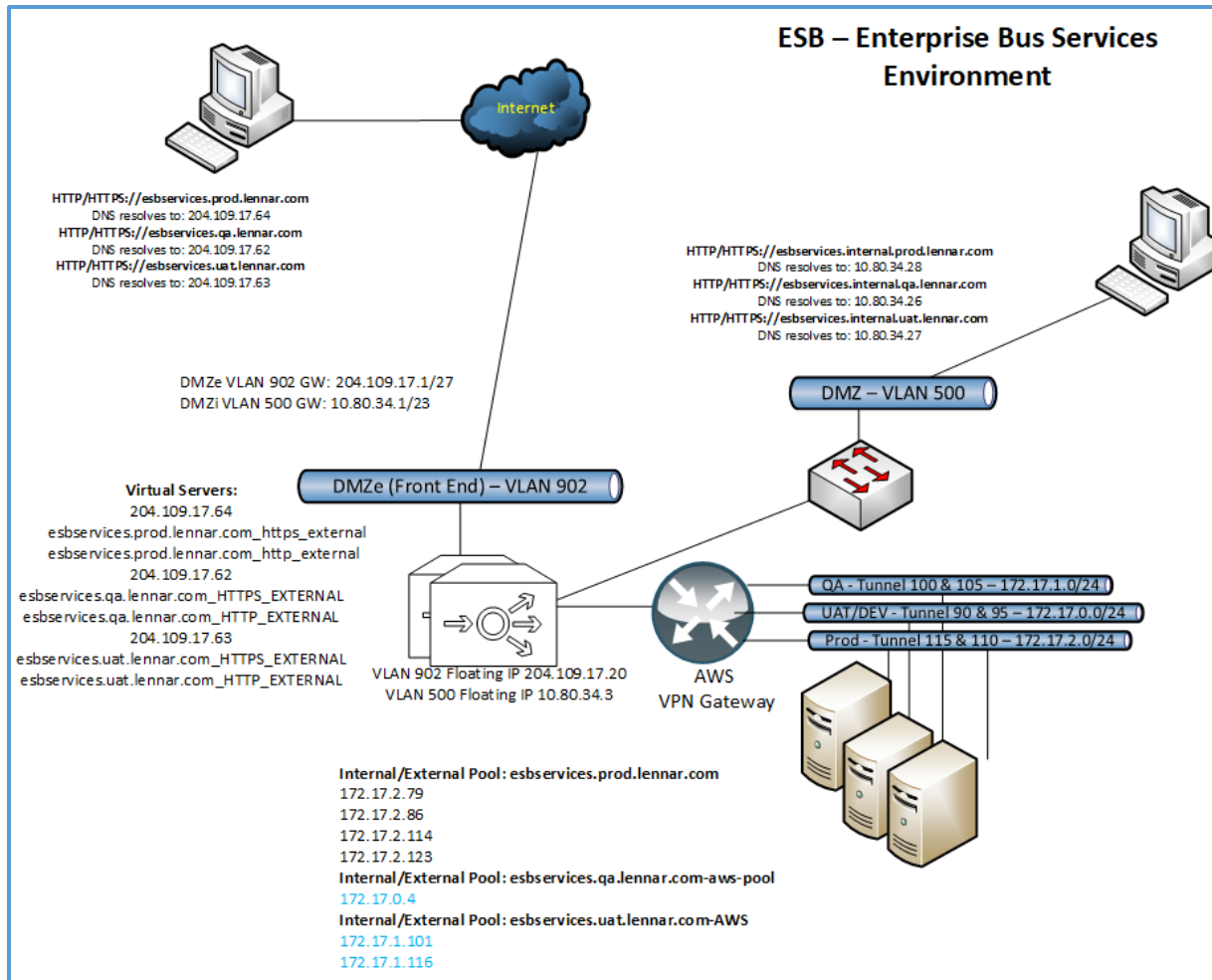


Figure 7: Enterprise Bus Services Environment

Micro Websites

The below logical diagram reflects the Lennar micro-websites. The micro-websites are http and https applications. The Docportal Apache server is a PHP based application used to make CURL calls to other applications outside of the Lennar.com domain. In addition, the lennaragentvip application makes API request to docportal application for a box refresh token.

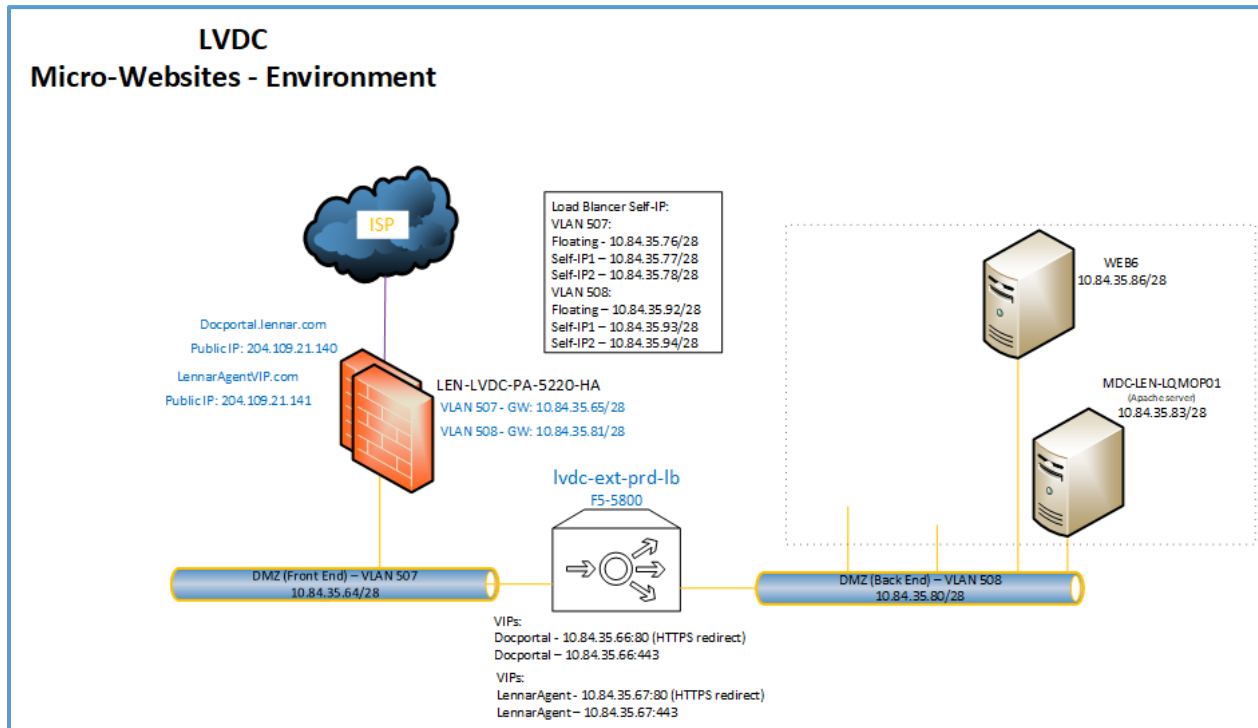


Figure 8: Micro-websites

INTERNAL

LDAP

The below diagram reflects the existing LDAP application in MDC. This application is used for directory services authentication. This application uses the service ports 389, 443, & 636.

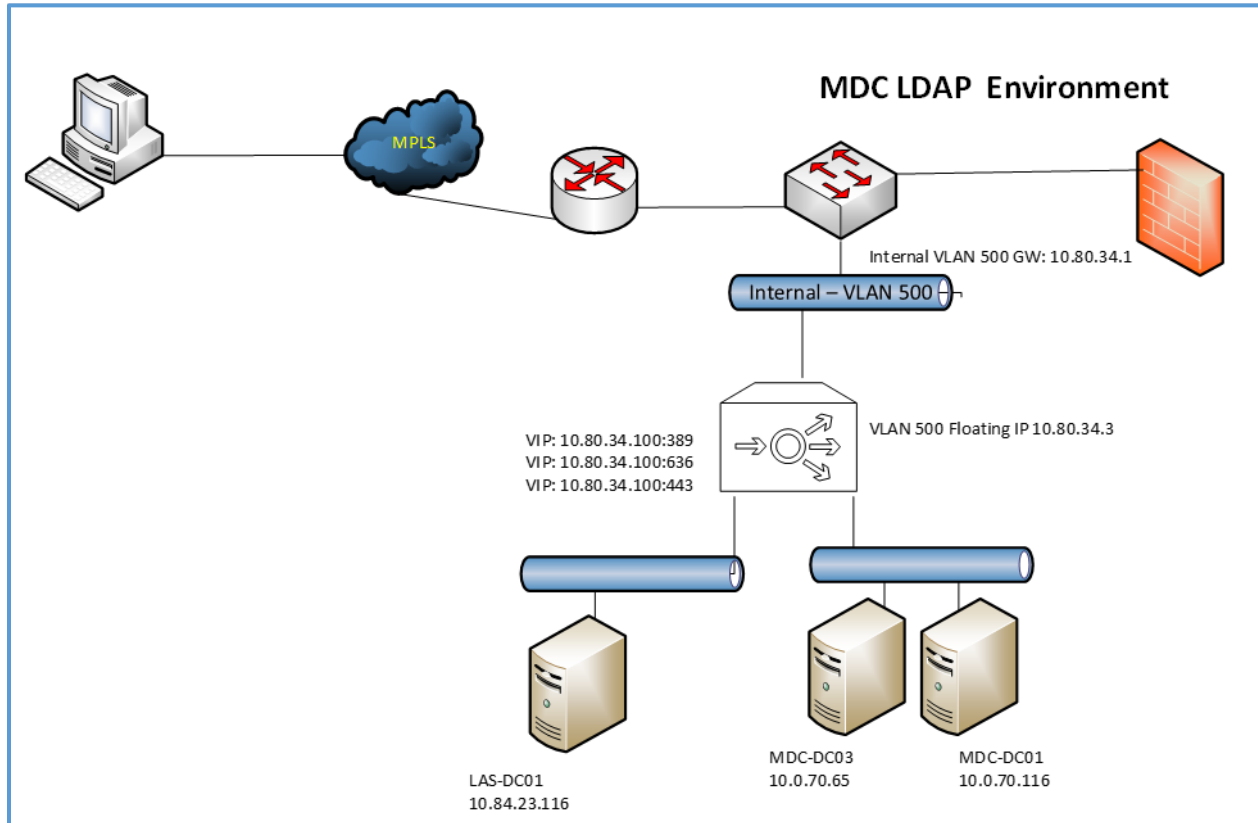


Figure 9: MDC LDAP Environment

DEV/UAT ENVIRONMENT

Lennar.com Environment (Dev)

The below diagram reflects the Lennar.com staging environment. This environment is used to test functionality and operability of the Lennar.com application prior to production. The development uses this environment to conduct thorough UAT and QA testing of this application. This application is presented to the client via HTTPS with redirects presented via HTTP to HTTPS for optimization.

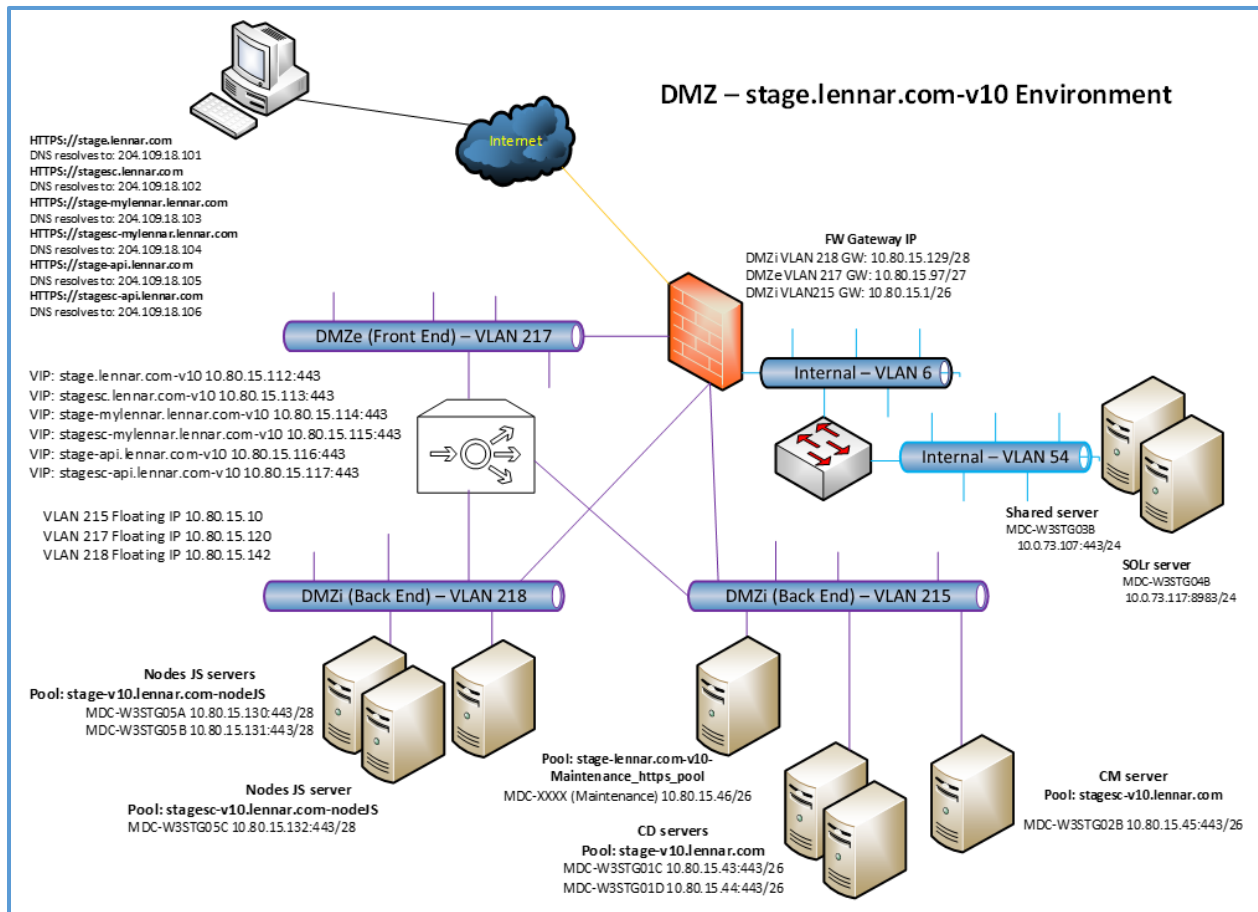


Figure 10: Lennar.com Environment